User Manual

Introduction

1.1 Purpose of the System

The TA Recruitment Management System is designed to support the Teaching Assistant recruitment process for universities. The system provides an online platform where TA applicants, Module Organisers, and System Administrators can complete different recruitment tasks according to their roles.

The main purpose of the system is to improve the efficiency of TA recruitment by reducing manual work and making the recruitment process easier to manage. Instead of relying mainly on emails and Excel files, users can complete the whole process through the system, including applicant registration, profile completion, CV upload, job browsing, application submission, application review, job publishing, workload monitoring, and administrative supervision.

For TA applicants, the system allows them to create an account, complete their personal profile, upload a CV, browse published TA jobs, use AI-assisted recommendations, submit applications, and track application status. For Module Organisers, the system supports reviewing applicant profiles and CVs, approving or rejecting applications, and publishing TA job vacancies. For System Administrators, the system provides dashboard statistics, workload monitoring, user management, job inspection, CSV export, and AI demo functions.

Overall, the system aims to provide a more organised, transparent, and efficient TA recruitment workflow. It also combines role-based access control and AI-assisted recruitment features to help users make better decisions during the recruitment process.

1.2 Purpose of the User Manual

This user manual provides guidance on how to use the TA Recruitment Management System. It is written for the main users of the system, including TA applicants, Module Organisers, and System Administrators.

The purpose of this manual is to explain the system functions, user roles, operation steps, expected results, and important rules in a clear and practical way. By following this manual, users should be able to understand how to access the system, log in with the correct role, complete their required tasks, and handle common situations during the recruitment process.

This manual follows the logical order of the system workflow. It first introduces the TA applicant process, including registration, profile completion, job browsing, AI-assisted analysis, and application submission. It then explains the Module Organiser workflow, including dashboard use, application review, and job publishing. Finally, it describes the Administrator workflow, including system monitoring, workload control, user management, job management, data export, TA profile overview, and AI demo functions.

This manual focuses on how users operate the system. It does not explain source code implementation details. Therefore, it can be used as a practical guide for demonstrations, system testing, and real user operation.

1.3 System Requirements

The TA Recruitment Management System is a web-based application. Users need to access the system through a web browser after the system has been deployed or started on a local server.

For normal users, such as TA applicants, Module Organisers, and System Administrators, the following requirements are needed:

Requirement	Description
Device	A computer or laptop
Web Browser	A modern browser such as Google Chrome, Microsoft Edge, or Firefox
Network / Server Access	Access to the local server or deployed system URL
User Account	A valid account with the correct role, such as TA, MO, or Admin

For running or deploying the system, the following software environment is required:

Tool / Component	Requirement
Java Development Kit	JDK 11 or above
Build Tool	Apache Maven 3.9 or above

Tool / Component	Requirement
Web Server	Apache Tomcat 9.x
Web Technology	Java Servlet 4.0 and JSP 2.3
Data Storage	Plain JSON files, no database required

The system can be run locally using embedded Tomcat. In this case, users can open the system through the local address:

http://localhost:18080/ta-recruitment/login

If the system is deployed to an external Tomcat server, users can access it through a Tomcat deployment URL, for example:

http://localhost:8080/ta-recruitment/login

The system stores user accounts, applicant profiles, job records, application records, and uploaded CV files on the server side using JSON files. Therefore, no separate database installation is required for this coursework prototype. The AI functions use a mock provider by default, so no external AI API key is required for the basic demonstration.

1.4 User Roles

The TA Recruitment Management System supports three main user roles: TA Applicant, Module Organiser, and System Administrator. Each role has different permissions and is redirected to a different main page after login. This role-based design ensures that users can only access the functions that are relevant to their responsibilities.

User Role	Description	Main Page After Login
TA Applicant	A student user who can register, complete a profile, browse TA jobs, apply for positions, use AI-assisted recommendations, and track application status.	/job
Module Organiser	A module organiser who can review TA applications, open applicant CVs, view applicant profiles, approve or reject applications, and create or publish TA jobs.	/mo
System Administrator	An administrator who can supervise the whole system, monitor statistics, manage users, inspect jobs, export data, view TA profiles, and	/admin

User Role	Description	Main Page After Login
	use the AI demo page.	

TA Applicants mainly use the system to complete the application process. They can register an account, fill in their personal profile, upload a CV, browse published jobs, view job details, use AI-assisted analysis, submit applications, and check their application results.

Module Organisers mainly use the system to manage recruitment tasks related to TA applications and job postings. They can review incoming applications, check applicants' CVs and profiles, provide feedback, approve or reject applications, and publish new TA job opportunities.

System Administrators have the highest level of supervision. They can monitor the whole platform through the administrator dashboard, check recruitment statistics, monitor TA workload, manage user accounts, inspect job records, export CSV data, browse TA profiles, and test AI functions through the AI demo page

2.1 User Registration

Purpose

The user registration function allows a new TA applicant to create an account in the system. Newly registered users are assigned the TA Applicant role by default.

Steps

1. Open the registration page.

2. Enter the required personal information, including name, student ID,
3. email address, phone number, password, and password confirmation.

4. Check that all information is entered in the correct format.

5. Click the Register button.

If the information is valid, the system creates a new TA applicant account.

6. After successful registration, the user is automatically logged in and redirected to the job list page.

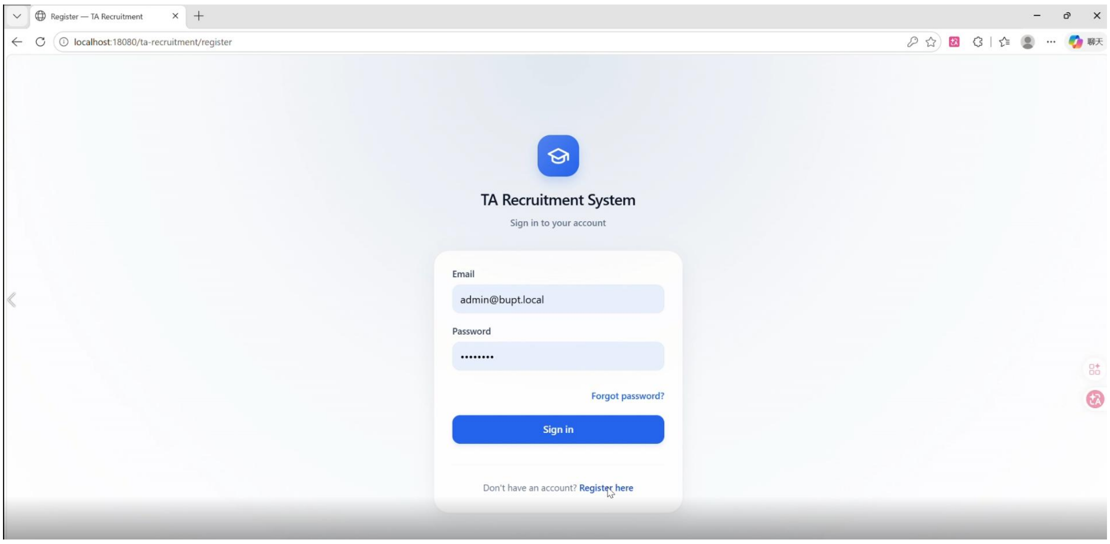
Validation Rules

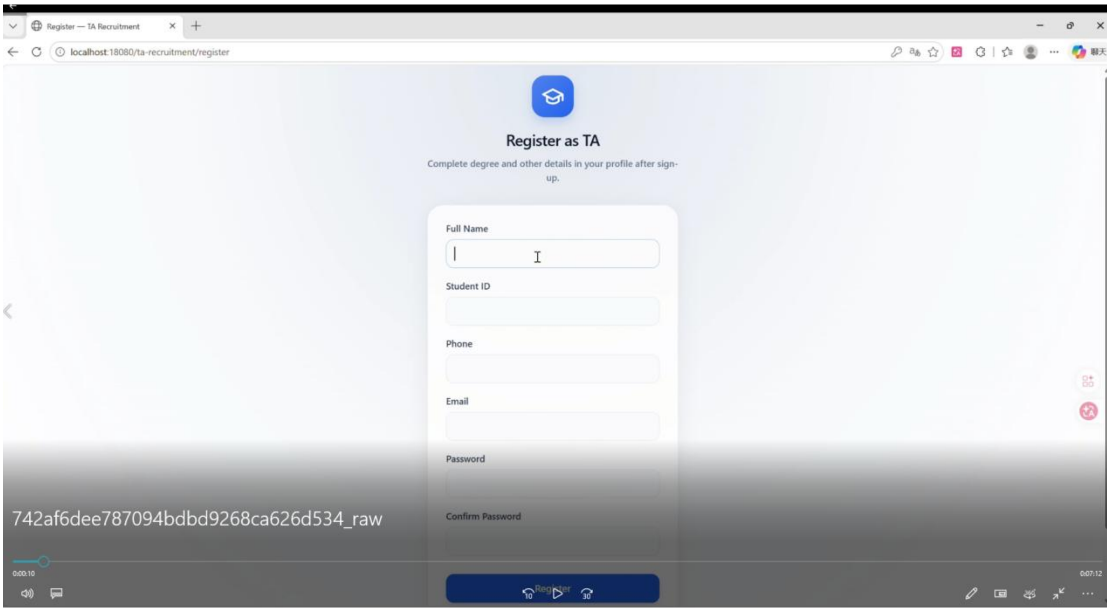
The system validates the registration information before creating the account. For example, the student ID, email format, phone number, password, and password confirmation must meet the required format. If any information is invalid, the system displays an error message and asks the user to correct it.

Expected Result

After successful registration, the TA applicant account is created. The system also creates an empty profile structure for the new user, so the applicant can complete the personal profile later.

Screenshot





2.2 Complete Personal Profile

Purpose

Before applying for TA jobs, applicants need to complete their personal profile. The profile provides important information for Module Organisers and Administrators when reviewing applications.

Profile Information

The profile page includes the following information:

Section	Description
Personal information	Basic details such as name, gender, degree, major, student ID, national ID, phone number, and email
Education history	The applicant’s education background; multiple records can be added
Completed courses	Courses the applicant has completed
Available working time	The applicant’s available time for TA work
Skills	The applicant’s skills; this field is also used by the AI functions
CV upload	The applicant can upload a CV file

Steps

1. Open the profile page.
2. Fill in the required personal information.
3. Add education history records.
4. Enter completed courses.
5. Enter available working time.
6. Enter relevant skills.
7. Upload a CV file.
8. Click **Save** to submit the profile information.

Rules and Restrictions

The applicant must complete the required profile information before applying for a job. If the profile is incomplete, the system redirects the applicant back to the profile page and asks them to complete the missing information first.

Once the profile is complete, the page becomes read-only to prevent accidental changes. If the applicant wants to edit the profile later, they need to enter edit mode using a special URL parameter.

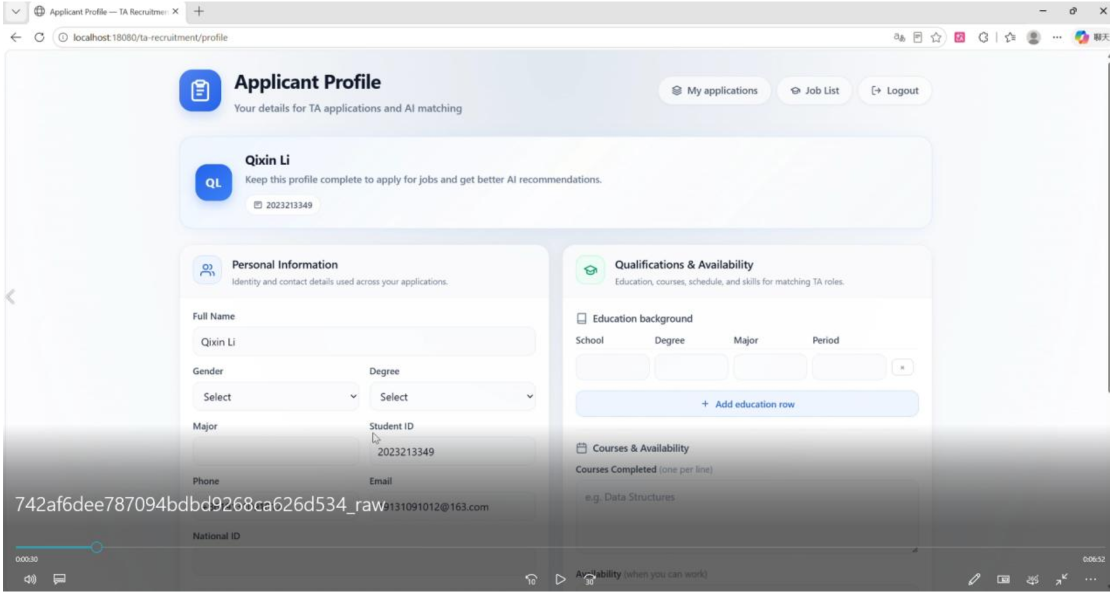
```
/profile?edit=1
```

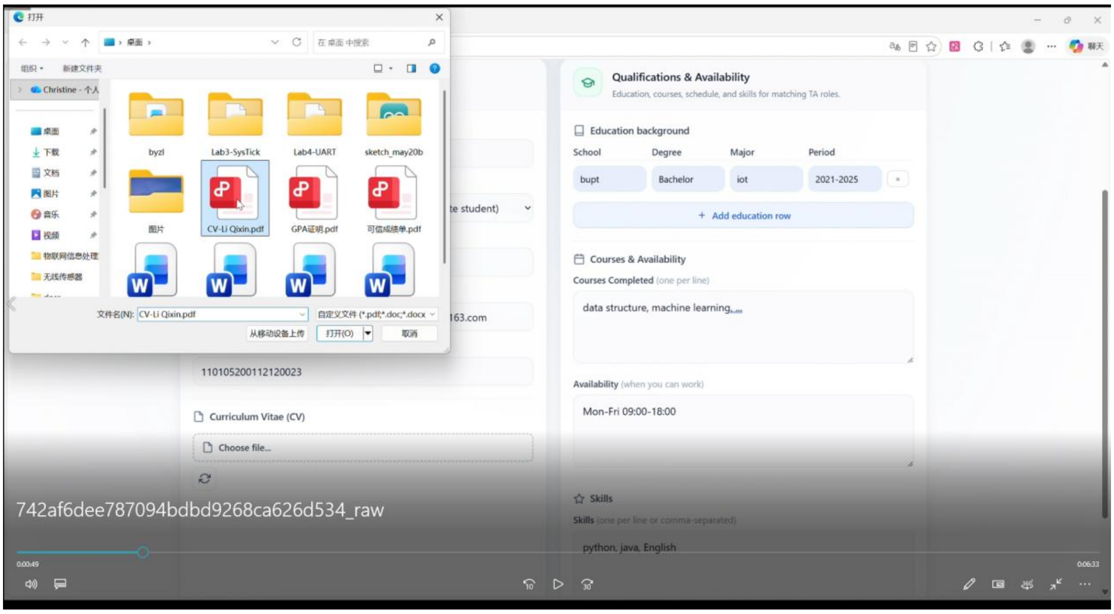
The CV upload function supports common document formats such as PDF, DOC, and DOCX. The uploaded CV can be opened or downloaded later.

Expected Result

The applicant's profile is saved successfully. A complete profile allows the applicant to apply for TA jobs and use AI-assisted functions.

Screenshot





2.3 Browse Jobs

Purpose

The job browsing function allows TA applicants to view available TA positions. Only published jobs are visible to TA applicants.

Main Functions

The job list page provides several functions to help applicants find suitable TA positions.

Function	Description
Job browsing	View all published TA jobs
Keyword search	Search jobs by module name, module description, or related keywords
Sorting	Sort jobs by module name, posting date, activity type, or favorited jobs
Favorites	Save jobs into the favorites list
Recently viewed jobs	Display jobs that the applicant has recently opened

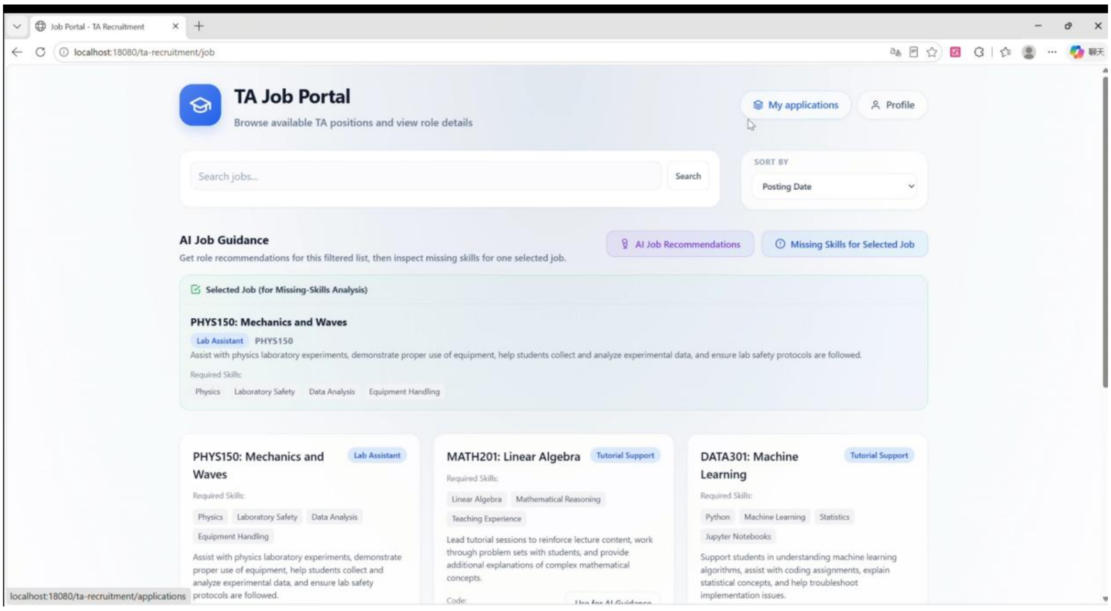
Steps

1. Open the job list page.
2. Browse the available published TA jobs.
3. Use the search box to find a specific job if needed.
4. Use sorting options to organise the job list.
5. Click **Save Job** or the favorite button to save a job.
6. Click a job from the list to open the job detail page.

Expected Result

The applicant can view available TA jobs, search for suitable positions, save interesting jobs, and open job details for further information.

Screenshot



2.4 Job Details

Purpose

The job detail page allows applicants to view detailed information about a selected TA position before applying.

Access

/job?id=jobId

Information Displayed

The job detail page shows important job information, including:

Information	Description
Module information	Module name, module code, and related details
Required skills	Skills expected for the TA position
Workload	Expected workload for the position
Deadline	Application deadline
TA quota	Number of available TA places
Application statistic	Current application status statistics for the job

Steps

1. Open the job list page.
2. Click a job to view its details.
3. Read the module information, required skills, workload, deadline, and TA quota.
4. Check the application statistics to understand the competition level.
5. Save the job to favorites if needed.
6. Click **Apply for Job** if the applicant meets the requirements.

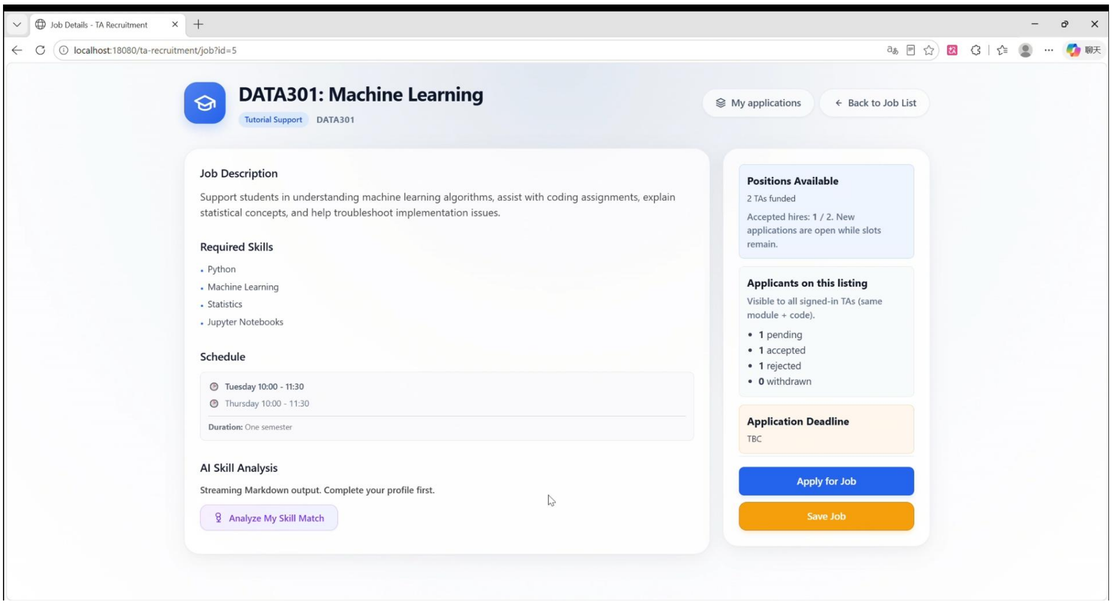
Rules and Restrictions

If the job ID is invalid, the system displays a **Job Not Found** message. If the TA quota is already full or the applicant does not meet the application rules, the apply button may be disabled and the system will show a related message.

Expected Result

The applicant can clearly understand the job requirements and decide whether to apply for the position.

Screenshot



2.5 AI-Assisted Recommendation and Analysis

Purpose

The system provides AI-assisted functions to help TA applicants make better decisions when choosing TA jobs. The AI functions are connected with the applicant's profile and the current job list. The AI output is only used as guidance and does not automatically submit applications or change application status.

2.5.1 AI Job Recommendation

Purpose

The AI Job Recommendation function provides personalised job suggestions based on the applicant's profile, skills, background, and the current visible job list.

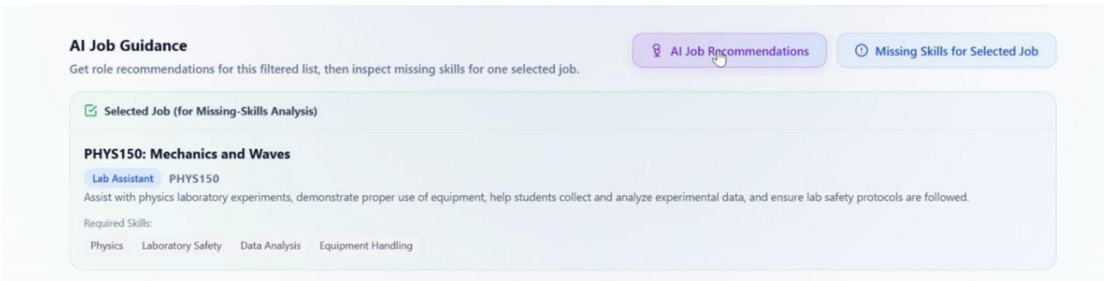
Steps

1. Open the job list page.
2. Make sure the profile information and skills have been completed.
3. Click **AI Job Recommendations**.
4. Wait for the AI response to appear progressively.
5. Read the recommended jobs and explanations.

Expected Result

The system provides personalised job recommendations to help the applicant choose suitable TA positions.

Screenshot



2.5.2 Missing Skills Analysis

Purpose

The Missing Skills Analysis function helps applicants identify the gap between their current skills and the skills required by a selected job.

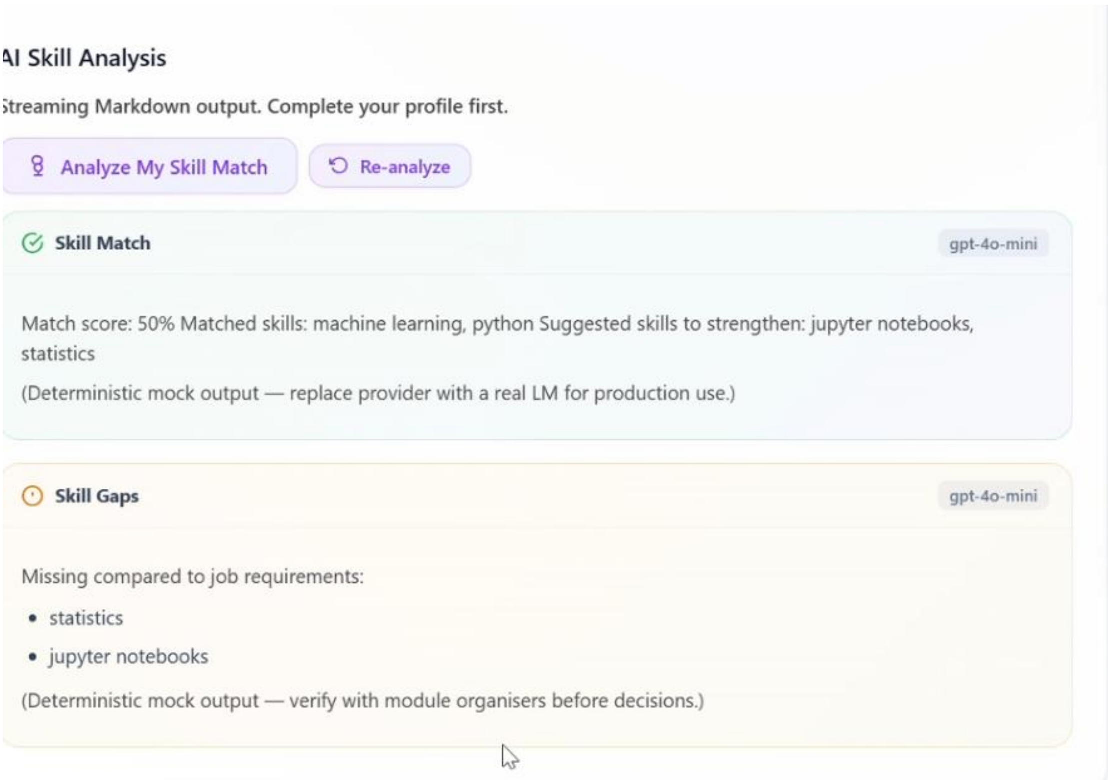
Steps

1. Open the job list page.
2. Select a specific job.
3. Click **Missing Skills Analysis** or **Missing Skills for Selected Job**.
4. Wait for the AI-generated analysis.
5. Read the missing skills and improvement suggestions.

Expected Result

The applicant can understand which skills are missing and what areas should be improved before applying for the job.

Screenshot



2.5.3 Skill Match Analysis

Purpose

The Skill Match Analysis function analyses how well the applicant's skills match a specific TA job.

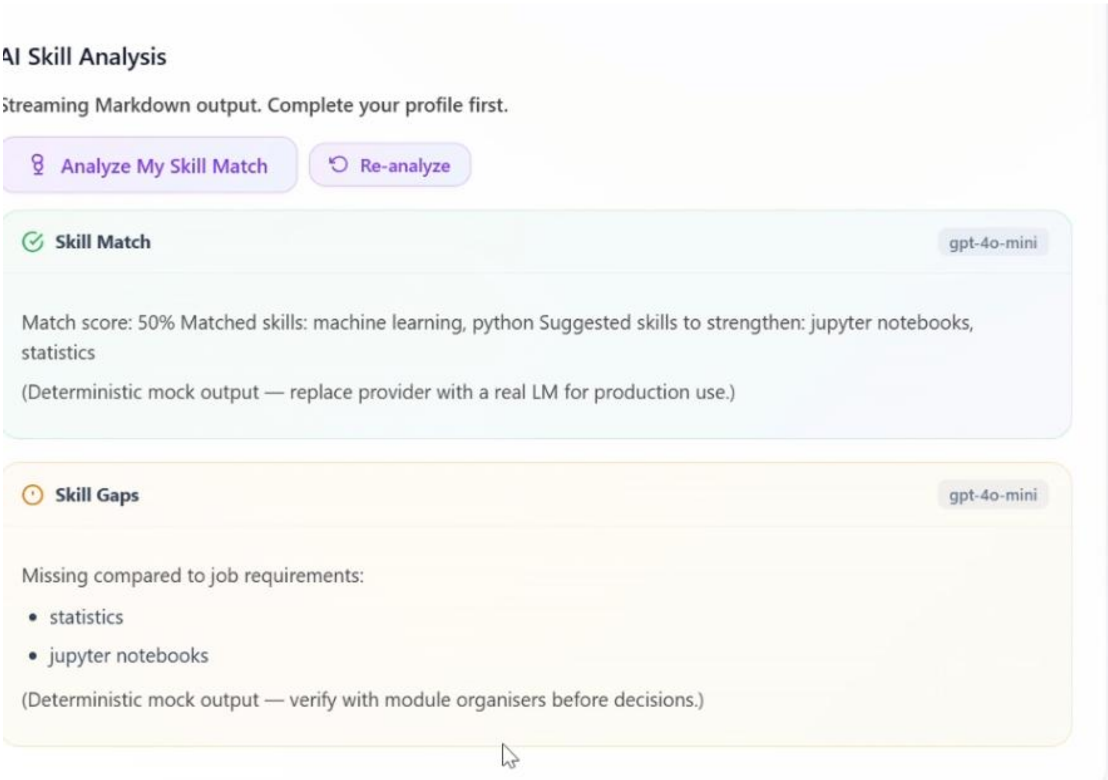
Steps

1. Open a job detail page.
2. Click **Analyze My Skill Match**.
3. Wait for the AI response.
4. Read the skill matching analysis and skill gap analysis.
5. Click **Re-analyze** if the applicant wants to generate the analysis again.

Expected Result

The applicant receives a skill matching result and a skill gap analysis, which can help them decide whether the job is suitable.

Screenshot



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2.6 Submit Application and My Applications

Purpose

This function allows TA applicants to submit applications for TA jobs and track their application status.

Access

`/applications`

Submit an Application

Steps

1. Open the job detail page.
2. Check the job information and requirements.
3. Click **Apply for Job**.
4. If the application is submitted successfully, the system redirects the applicant to the **My Applications** page.

Expected Result

The application is submitted and displayed in the applicant's application list.

My Applications

The **My Applications** page allows applicants to manage and track their submitted applications.

Function	Description
Track application status	View whether an application is Pending, Accepted, Rejected, or Withdrawn
View feedback	Read feedback provided by the MO or Admin
Filter applications	Filter applications by status
Withdraw application	Withdraw an application if it is still Pending

Rules and Restrictions

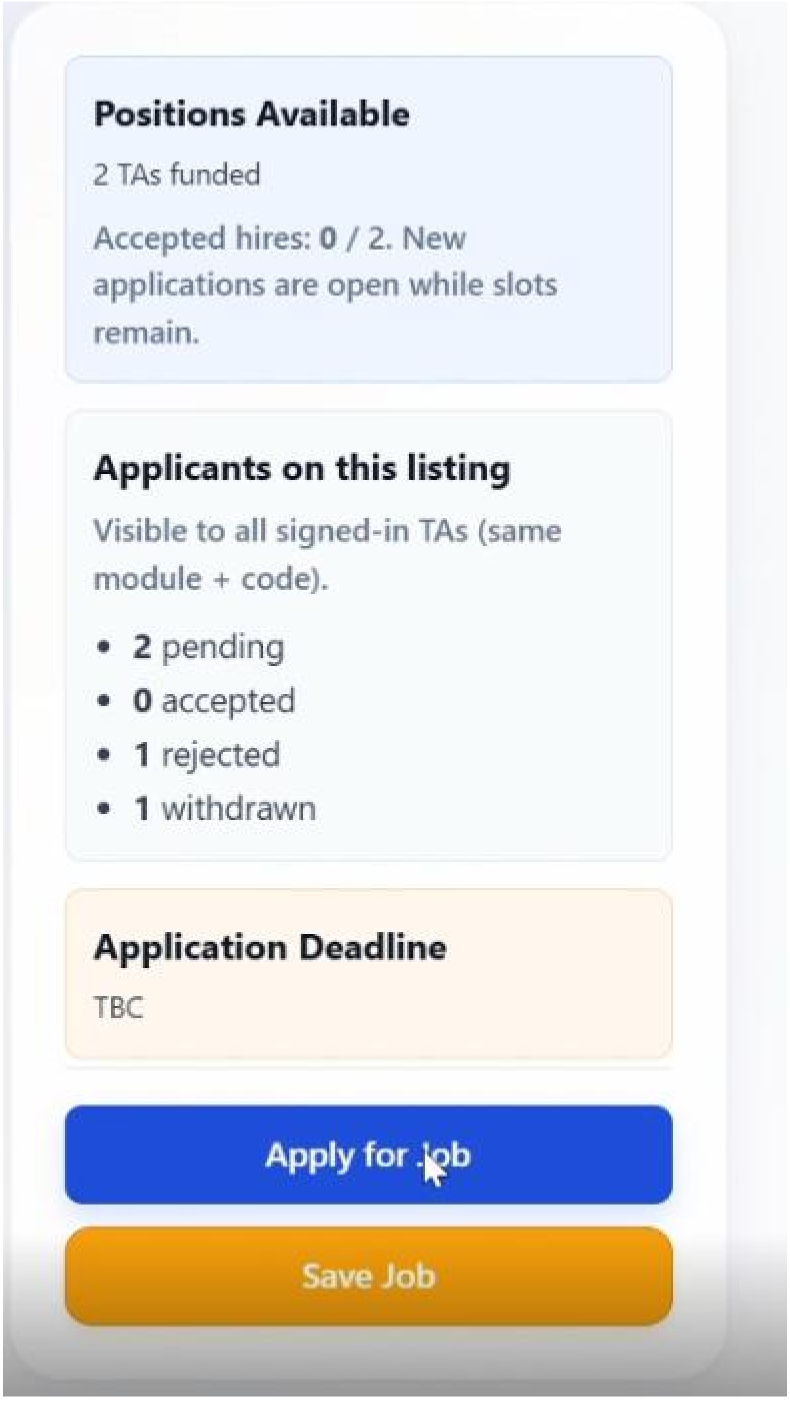
Only applications with **Pending** status can be withdrawn. Applications that have already been **Accepted** or **Rejected** cannot be withdrawn.

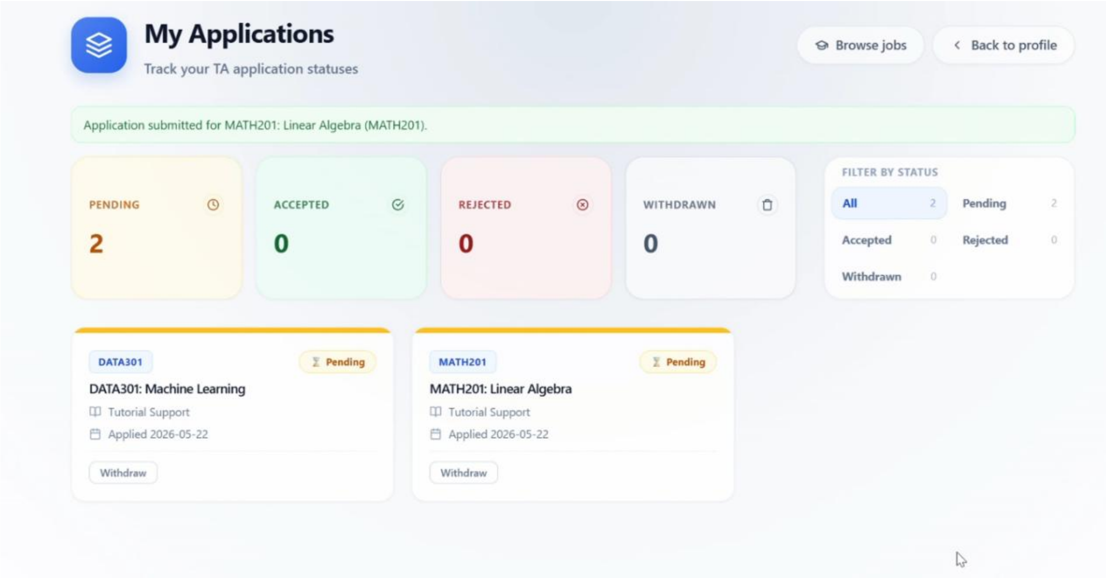
If a Module Organiser approves or rejects an application, the application status is updated. The applicant can refresh the **My Applications** page to view the latest status.

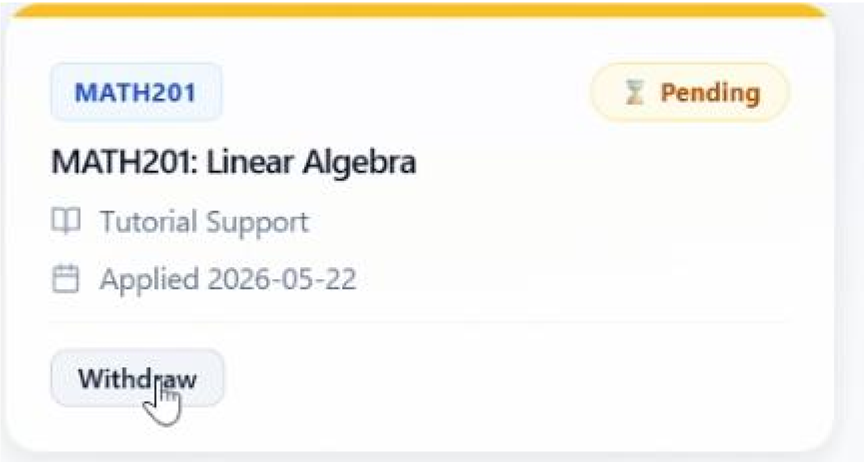
Expected Result

The applicant can clearly track all submitted applications and understand the current recruitment decision for each application.

Screenshot


Positions Available
2 TAs funded
Accepted hires: 0 / 2. New applications are open while slots remain.
Applicants on this listing
Visible to all signed-in TAs (same module + code).
2 pending
0 accepted
1 rejected
1 withdrawn
Application Deadline
TBC
Apply for Job
Save Job





3. Module Organiser Workflow

This section explains how Module Organisers use the system to manage TA recruitment tasks. A Module Organiser can view the MO dashboard, review incoming applications, open applicant CVs, view applicant profiles, approve or reject applications, provide feedback, and create or publish TA job postings.

After logging in as a Module Organiser, the user is redirected to the MO dashboard. The MO section is mainly used for reviewing TA applications and managing TA job postings.

3.1 MO Dashboard

Purpose

The MO Dashboard provides Module Organisers with an overview of recruitment information. It helps MO users quickly check job statistics, application statistics, and pending applications.

Access

/mo

Main Information Displayed

The MO dashboard displays key information related to jobs and applications.

Dashboard Item	Description
Total jobs	Shows the total number of TA job postings in the system
Total applications	Shows the total number of submitted TA applications
Pending applications	Shows the number of applications that are still waiting for review
Incoming Applications	Allows MO users to review submitted applications
Job Management	Allows MO users to manage job postings
Create / Publish Job Form	Allows MO users to create new TA jobs or publish them

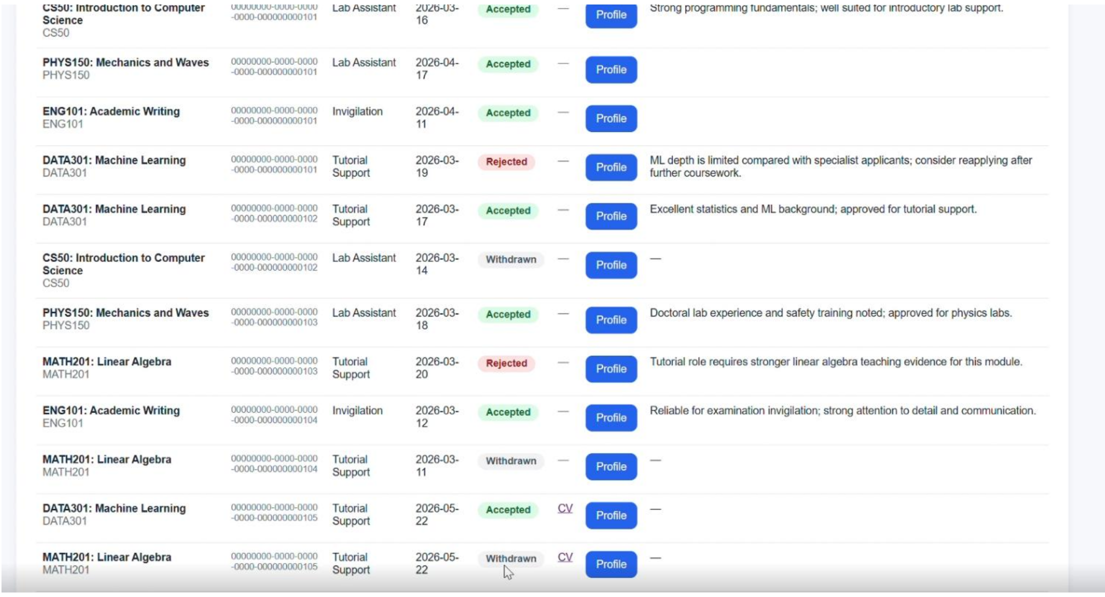
Steps

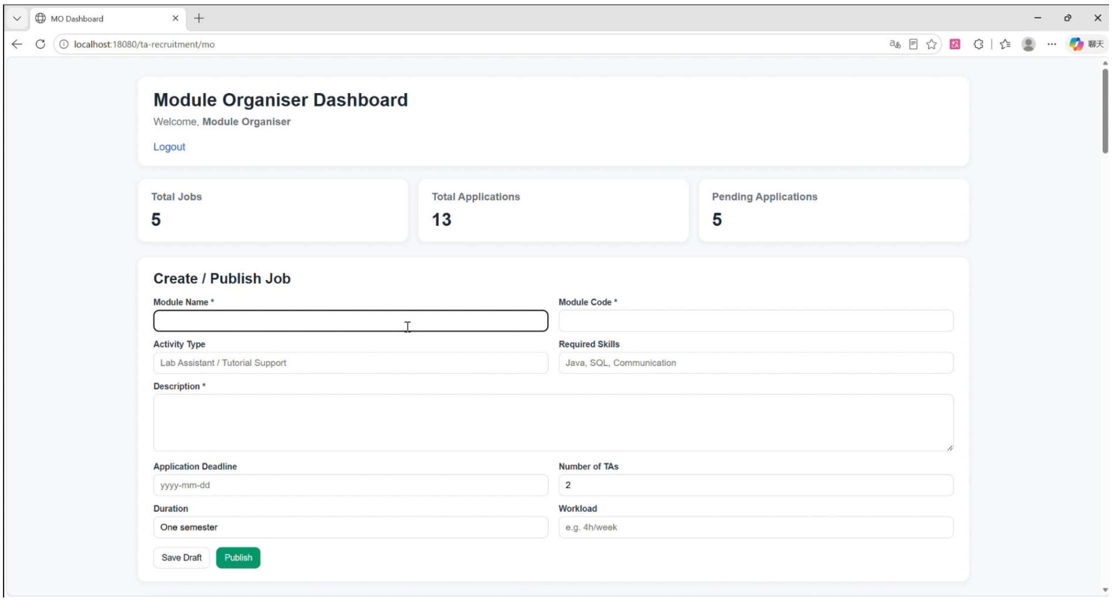
1. Log in with a Module Organiser account.
2. The system redirects the user to the MO dashboard.
3. Check the statistics cards, such as total jobs, total applications, and pending applications.
4. Scroll to the **Incoming Applications** section to review submitted applications.
5. Use the **Job Management** section to check existing jobs.
6. Use the job creation form to create or publish a new TA job.

Expected Result

The Module Organiser can clearly view the current recruitment situation and access the main MO functions from one page.

Screenshot





3.2 Review Applications

Purpose

The application review function allows Module Organisers to review TA applications and make recruitment decisions. MO users can check applicant information, open uploaded CV files, view applicant profiles, and approve or reject pending applications.

The application review function is located in the **Incoming Applications** section of the MO dashboard.

Main Functions

Function	Description
Open applicant CV	Opens the applicant’ s uploaded CV in a new window
View applicant profile	Opens the applicant’ s profile in read-only mode
Approve application	Changes a Pending application to Accepted
Reject application	Changes a Pending application to Rejected
Add feedback	Allows the MO to provide feedback when processing an application
Processed status	Shows that an application has already been handled

The system provides an MO-only read-only applicant profile page through `/mo/applicant-profile?userId=...`, and the README also lists `/mo` as the Module Organiser dashboard route.

Steps

1. Log in as a Module Organiser.
2. Open the MO dashboard.
3. Find the **Incoming Applications** section.
4. Select an application from the list.
5. Click **Open CV** to review the applicant’s uploaded CV if needed.
6. Click **View Profile** to check the applicant’s profile information.
7. If the applicant is suitable, click **Approve**.
8. If the applicant is not suitable, click **Reject**.
9. Enter feedback if necessary.
10. Submit the decision.

Rules and Restrictions

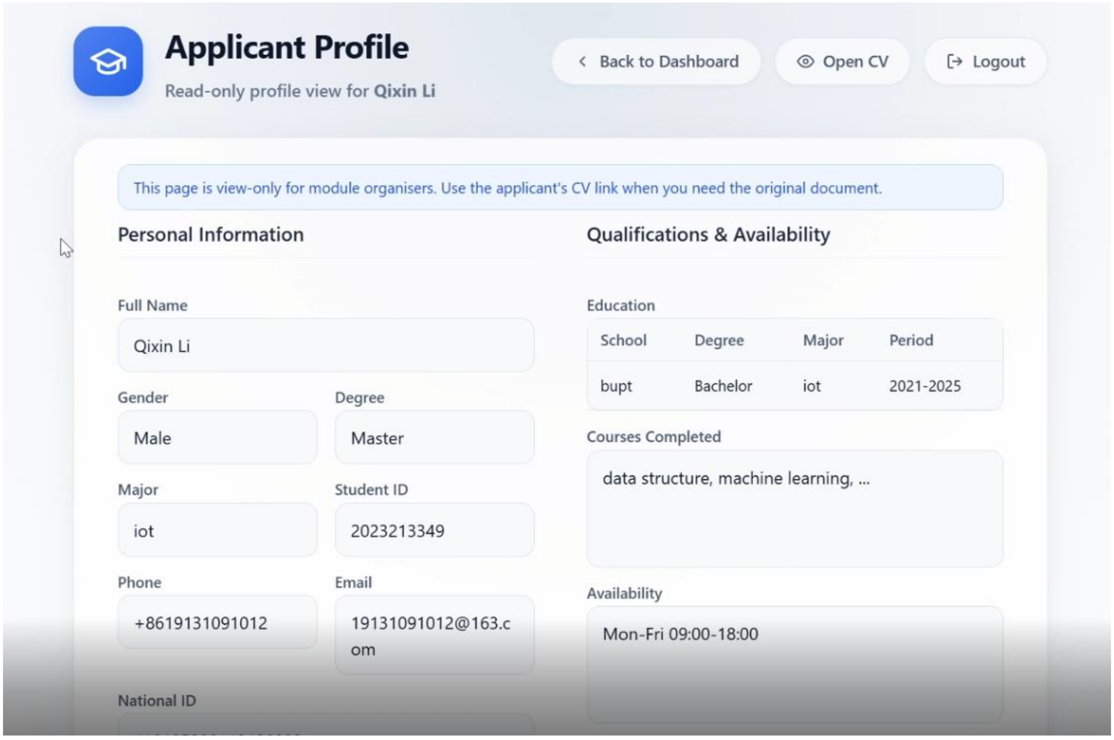
Only applications with **Pending** status can be approved or rejected by the MO. Once an application has been processed, it is marked as **Processed** and cannot be approved or rejected again.

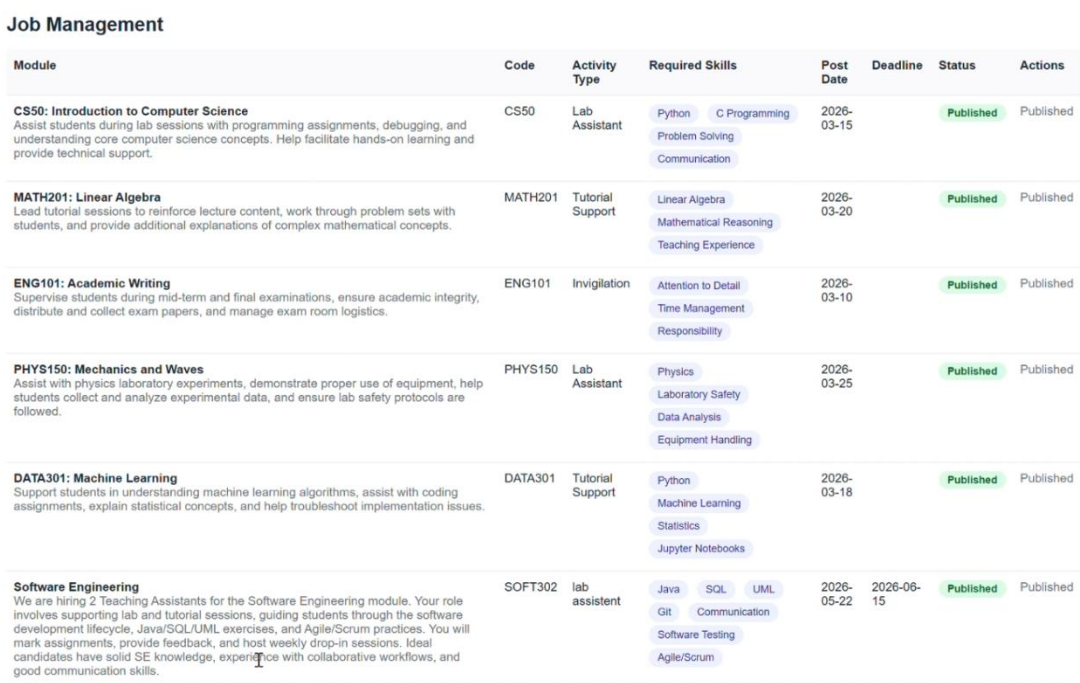
When the MO approves or rejects an application, the applicant can see the updated status on the **My Applications** page after refreshing. This shows the workflow connection between the TA applicant side and the MO side.

Expected Result

The selected application status is updated from **Pending** to **Accepted** or **Rejected**. If feedback is provided, the applicant can view the feedback in their application page.

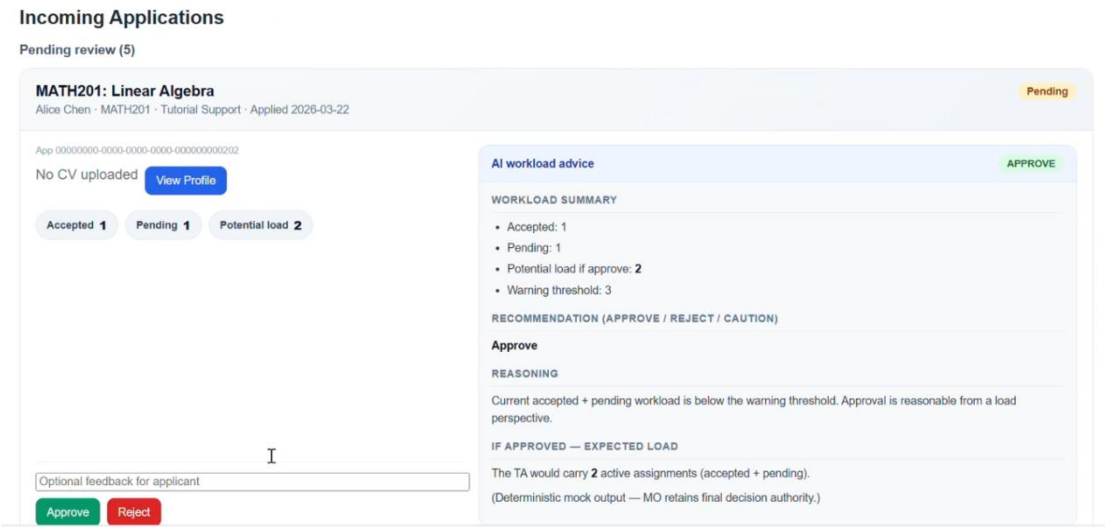
Screenshot





Job Management

Module	Code	Activity Type	Required Skills	Post Date	Deadline	Status	Actions
CS50: Introduction to Computer Science Assist students during lab sessions with programming assignments, debugging, and understanding core computer science concepts. Help facilitate hands-on learning and provide technical support.	CS50	Lab Assistant	Python, C Programming, Problem Solving, Communication	2026-03-15		Published	Published
MATH201: Linear Algebra Lead tutorial sessions to reinforce lecture content, work through problem sets with students, and provide additional explanations of complex mathematical concepts.	MATH201	Tutorial Support	Linear Algebra, Mathematical Reasoning, Teaching Experience	2026-03-20		Published	Published
ENG101: Academic Writing Supervise students during mid-term and final examinations, ensure academic integrity, distribute and collect exam papers, and manage exam room logistics.	ENG101	Invigilation	Attention to Detail, Time Management, Responsibility	2026-03-10		Published	Published
PHYS150: Mechanics and Waves Assist with physics laboratory experiments, demonstrate proper use of equipment, help students collect and analyze experimental data, and ensure lab safety protocols are followed.	PHYS150	Lab Assistant	Physics, Laboratory Safety, Data Analysis, Equipment Handling	2026-03-25		Published	Published
DATA301: Machine Learning Support students in understanding machine learning algorithms, assist with coding assignments, explain statistical concepts, and help troubleshoot implementation issues.	DATA301	Tutorial Support	Python, Machine Learning, Statistics, Jupyter Notebooks	2026-03-18		Published	Published
Software Engineering We are hiring 2 Teaching Assistants for the Software Engineering module. Your role involves supporting lab and tutorial sessions, guiding students through the software development lifecycle, Java/SQL/UML exercises, and Agile/Scrum practices. You will mark assignments, provide feedback, and host weekly drop-in sessions. Ideal candidates have solid SE knowledge, experience with collaborative workflows, and good communication skills.	SOFT302	lab assistent	Java, SQL, UML, Git, Communication, Software Testing, Agile/Scrum	2026-05-22	2026-06-15	Published	Published



3.3 Publish Jobs

Purpose

The job publishing function allows Module Organisers to create new TA job postings. MO users can save a job as a draft or publish it directly. Published jobs become visible to TA applicants on the job list page, while draft jobs remain invisible to applicants.

The job publishing function is available on the MO dashboard.

Required Job Information

When creating a new TA job, the Module Organiser needs to provide basic job information.

Information	Description
Module information	The module name, module code, or related module details
TA quota	The number of TA positions available
Workload	The expected workload for the TA position
Deadline	The application deadline
Required skills	The skills required for the TA role
Job description	Additional information about the job responsibilities

Available Actions

Action	Result
Save Draft	The job is saved as a draft and is not visible to TA applicants
Publish	The job is published and becomes visible to TA applicants immediately
Publish from Job Management	A draft job can be published later from the Job Management section

The feature map states that draft jobs are invisible to TA applicants, while published jobs appear immediately on the TA job list.

Steps

1. Log in as a Module Organiser.
2. Open the MO dashboard.
3. Find the job creation form.
4. Enter the required module information.
5. Enter the TA quota, workload, deadline, required skills, and job description.
6. Click **Save Draft** if the job should not be shown to applicants yet.
7. Click **Publish** if the job is ready to be shown to applicants.
8. If a job is saved as a draft, open the **Job Management** section later and click **Publish** when it is ready.

Rules and Restrictions

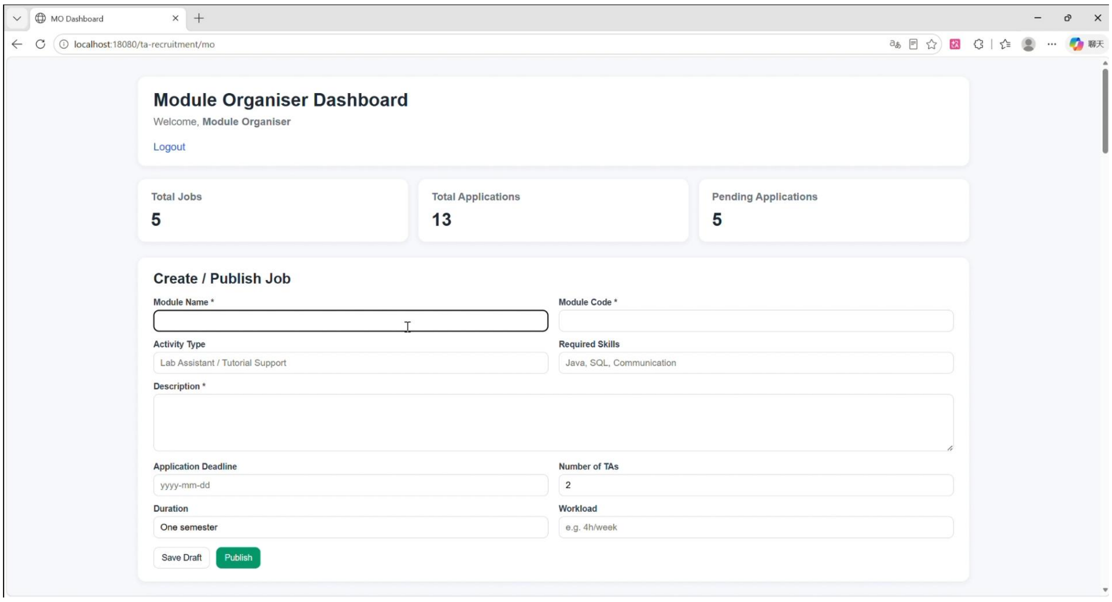
Draft jobs are not visible to TA applicants. Only published jobs can be viewed and applied for by TA applicants on the job list page.

The MO can create and publish jobs, but the MO does not have administrator-level permissions. For example, MO users cannot force application status changes, manage user accounts, delete jobs as an administrator, export CSV data, or use the Admin AI Demo page.

Expected Result

If the job is saved as a draft, it remains in the MO job management area and is hidden from TA applicants. If the job is published, it becomes visible on the TA job list page immediately.

Screenshot



4. Administrator Workflow

This section explains how System Administrators use the TA Recruitment Management System to supervise and manage the whole platform. Administrators have the highest level of system access. They can monitor recruitment statistics, manage users, inspect job records, export system data, review TA profiles, and use the AI demo page for testing.

4.1 Dashboard and Analytics

Purpose

The Administrator Dashboard allows administrators to monitor the overall status of the TA recruitment system. It provides statistics, charts, and status analysis to help administrators understand the current recruitment situation.

Main Information Displayed

Dashboard Item	Description
Total TA accounts	Shows the total number of registered TA applicant accounts
Total jobs	Shows the total number of TA job records, including

Dashboard Item	Description
	draft and published jobs
Total applications	Shows the total number of submitted applications
Application status analysis	Shows the number of Pending, Accepted, and Rejected applications
Monthly Application Trend	Displays application trends by month
Top Modules by Applications	Shows the modules with the highest number of applications
Workload monitoring	Helps administrators identify overloaded TAs

The administrator dashboard is used to monitor the platform through statistics and charts, including user statistics, job statistics, application analysis, monthly trends, and workload monitoring.

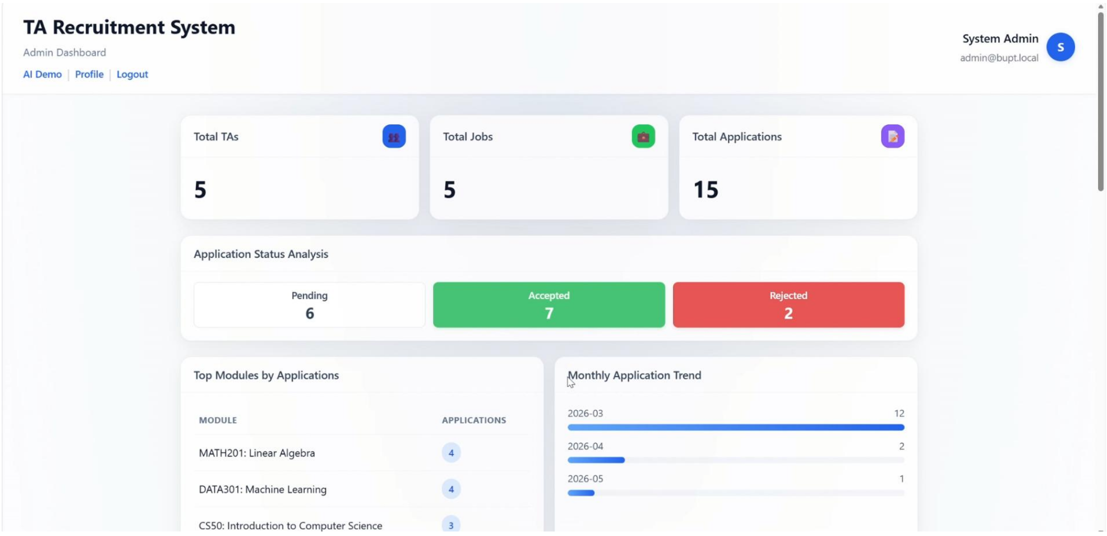
Steps

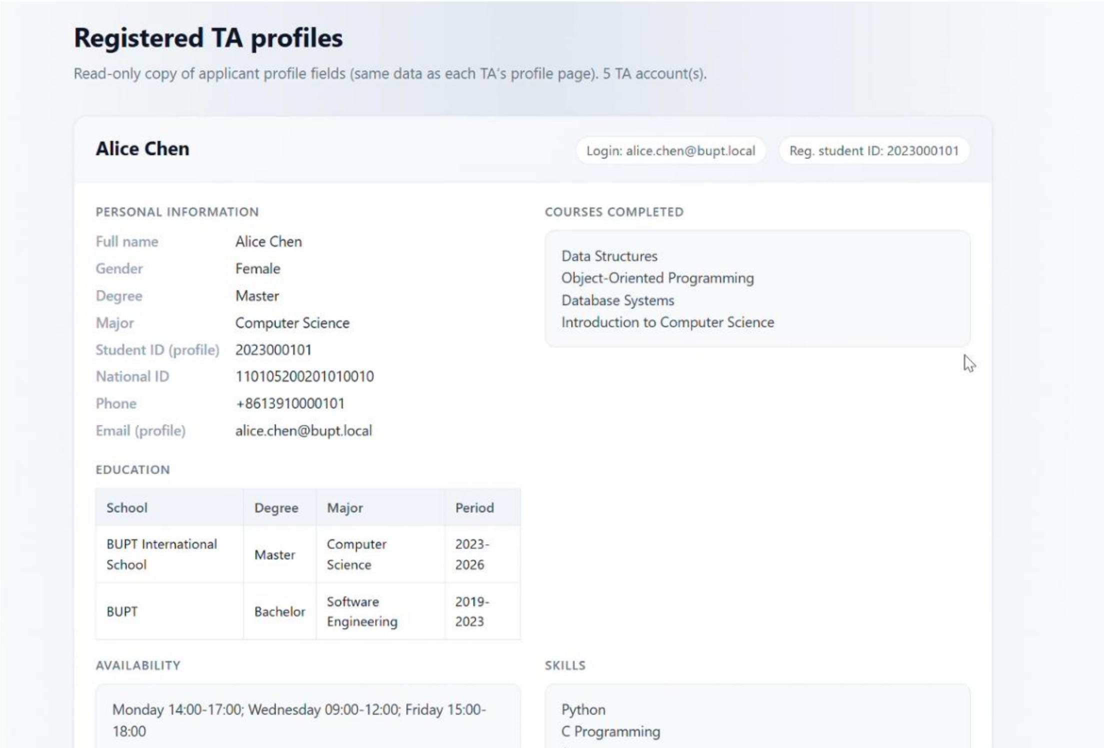
1. Log in with an Administrator account.
2. Open the Administrator Dashboard.
3. Check the overall statistics, such as total TA accounts, total jobs, and total applications.
4. Review the application status analysis.
5. Check the monthly application trend chart.
6. View the top modules by application number.
7. Use the status links to open detailed application lists if needed.

Expected Result

The administrator can clearly understand the overall recruitment situation and quickly identify important system information.

Screenshot





4.2 Workload Monitoring

Purpose

The Workload Monitoring function helps administrators check whether any TA has accepted too many positions. This supports better workload balance and prevents over-allocation of TA work.

Access

/admin

The workload monitoring section is displayed on the Administrator Dashboard.

Main Functions

Function	Description
TA Workload Overview	Shows TA workload information
Assigned jobs	Displays the number of accepted jobs assigned to a TA
Workload warning	Highlights TAs who may be overloaded
Accepted / Rejected job details	Shows related job information for each TA

Rules

The system automatically detects overloaded TAs. If a TA has accepted too many positions, the system shows a workload warning. In the feature map, overload is defined as a TA having **three or more Accepted jobs**.

Steps

1. Log in as an Administrator.
2. Open the Administrator Dashboard.
3. Find the **TA Workload Overview** section.
4. Check the assigned jobs for each TA.
5. Look for any workload warning.
6. Use the information to support workload balancing decisions.

Expected Result

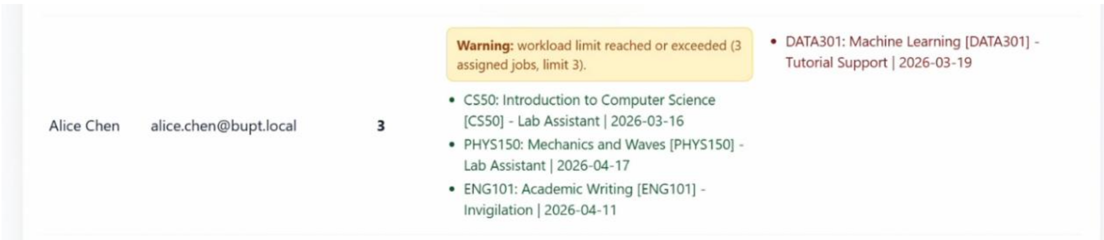
The administrator can identify overloaded TAs and make better decisions when supervising TA assignments.

Screenshot

TA Workload Overview

Only **TA** accounts. **Assigned jobs** = accepted applications. Warning when accepted ≥ 3.

TA NAME	EMAIL	ASSIGNED JOBS	ACCEPTED WORK	REJECTED WORK
Brian Li	brian.li@bupt.local	1	• DATA301: Machine Learning [DATA301] - Tutorial Support | 2026-03-17	—
Clara Wang	clara.wang@bupt.local	1	• PHYS150: Mechanics and Waves [PHYS150] - Lab Assistant | 2026-03-18	• MATH201: Linear Algebra [MATH201] - Tutorial Support | 2026-03-20
Daniel Zhang	daniel.zhang@bupt.local	1	• ENG101: Academic Writing [ENG101] - Invigilation | 2026-03-12	—
Qixin Li	19131091012@163.com	1	• DATA301: Machine Learning [DATA301] - Tutorial Support | 2026-05-22	—
Alice Chen	alice.chen@bupt.local	3	**Warning:** workload limit reached or exceeded (3 assigned jobs, limit 3). • CS50: Introduction to Computer Science [CS50] - Lab Assistant | 2026-03-16 • PHYS150: Mechanics and Waves [PHYS150] - Lab Assistant | 2026-04-17 • ENG101: Academic Writing [ENG101] - Invigilation | 2026-04-11	• DATA301: Machine Learning [DATA301] - Tutorial Support | 2026-03-19



Alice Chen	alice.chen@bupt.local	3	**Warning:** workload limit reached or exceeded (3 assigned jobs, limit 3). • CS50: Introduction to Computer Science [CS50] - Lab Assistant | 2026-03-16 • PHYS150: Mechanics and Waves [PHYS150] - Lab Assistant | 2026-04-17 • ENG101: Academic Writing [ENG101] - Invigilation | 2026-04-11	• DATA301: Machine Learning [DATA301] - Tutorial Support | 2026-03-19

4.3 Force Status Change

Purpose

The Force Status Change function allows administrators to directly override application statuses. This function provides a higher level of administrative control compared with the MO workflow.

Access

```
/admin/applications/by-status?status=pending
/admin/applications/by-status?status=accepted
/admin/applications/by-status?status=rejected
```

Available Actions

Current Status	Available Action	Result
Pending	Force accept	Changes status to Accepted
Pending	Force reject	Changes status to Rejected
Accepted	Force reject	Changes status to Rejected
Accepted	Force pend	Changes status back to Pending
Rejected	Force accept	Changes status to Accepted
Rejected	Force pend	Changes status back to Pending

Unlike MO users, administrators can directly override application statuses. For example, an Accepted application can be forced back to Pending or changed to Rejected.

Steps

1. Open the Administrator Dashboard.
2. Click an application status list, such as Pending, Accepted, or Rejected.
3. Find the application that needs to be updated.
4. Select the appropriate force status action.
5. Enter feedback if necessary.
6. Confirm the operation.
7. Check that the application status has been updated.

Rules and Restrictions

This function should be used carefully because it directly changes the application result. The action may require confirmation before execution.

Expected Result

The selected application status is updated immediately according to the administrator’s decision.

Screenshot

4.4 User Management

Purpose

The User Management function allows administrators to manage user accounts globally. Administrators can deactivate or delete user accounts when necessary.

Access

```
/admin
```

or

```
/admin/users
```

Available Actions

Action	Description
Deactivate	Deactivates a user account. The user can no longer log in.
Delete	Deletes a user account and related information.

Rules and Restrictions

The system prevents dangerous operations. Administrators cannot deactivate themselves, delete themselves, deactivate the only administrator account, or delete the only administrator account. A deactivated user can no longer log in to the system.

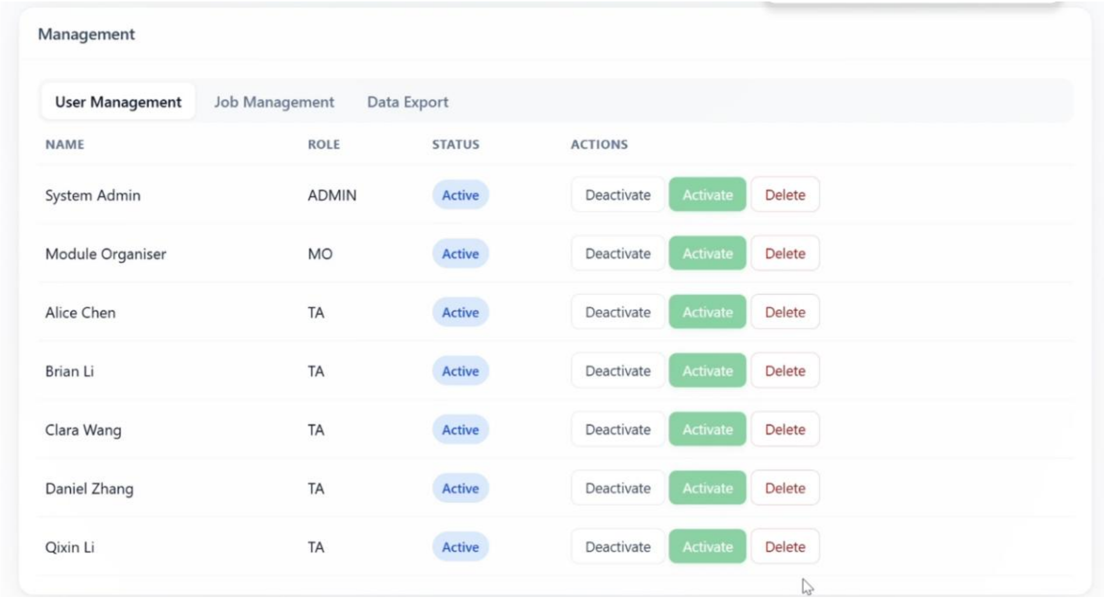
Steps

1. Log in as an Administrator.
2. Open the Administrator Dashboard.
3. Find the **User Management** section.
4. Select the user account that needs to be managed.
5. Click **Deactivate** if the user should no longer log in.
6. Click **Delete** if the user account should be removed.
7. Confirm the operation if required.

Expected Result

The selected user account is deactivated or deleted. If the account is deactivated, the user cannot log in anymore.

Screenshot



Insert screenshot of the User Management section here.

4.5 Job Management

Purpose

The Job Management function allows administrators to inspect and manage job records in the system. Administrators can view job information, open a read-only job detail page, and delete job records when necessary.

Access

`/admin/jobs`

Read-only job detail page:

`/admin/job-view?id=jobId`

Main Functions

Function	Description

Function	Description
View job list	Allows administrators to view existing job records
Inspect job details	Opens a read-only job detail page
Delete job	Removes a job record from the system
Confirmation before deletion	Prevents accidental job deletion

The system provides a read-only job detail page for administrators, allowing them to inspect job information without entering the MO workflow. Administrators can also remove job records, and deletion requires confirmation for safety.

Steps

1. Log in as an Administrator.
2. Open the **Job Management** section.
3. View the list of existing job records.
4. Click **View** to inspect job details in read-only mode.
5. If a job record needs to be removed, click **Delete**.
6. Confirm the deletion operation.

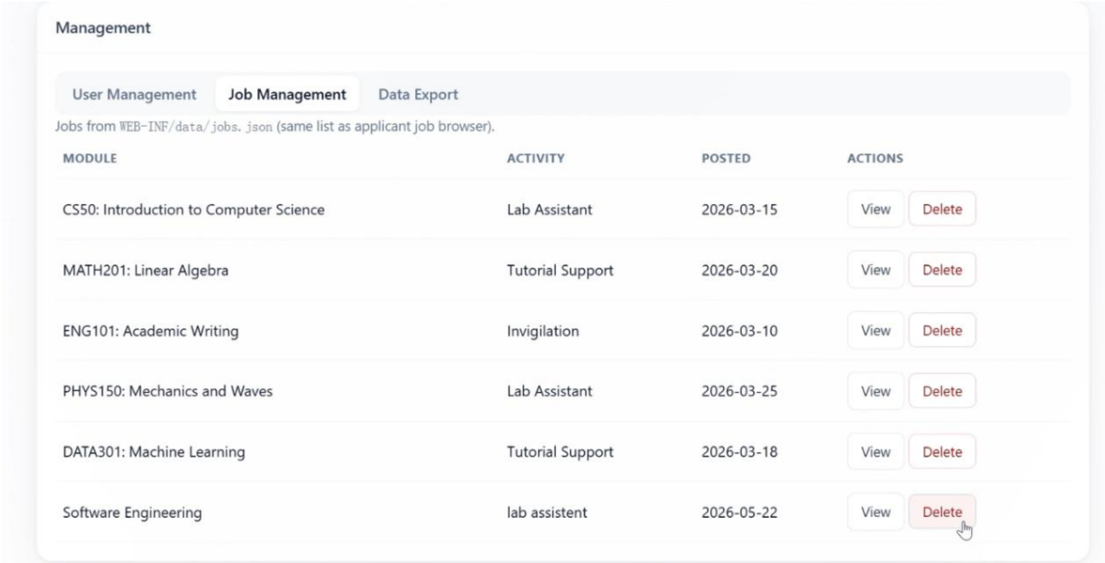
Rules and Restrictions

Administrators can inspect and delete job records, but they do not create or publish jobs from this page. Job creation and publishing are handled by Module Organisers in the MO workflow.

Expected Result

The administrator can inspect recruitment job records and remove unnecessary job records when required.

Screenshot



4.6 CSV Export

Purpose

The CSV Export function allows administrators to export system data for reporting and administrative analysis.

Access

`/admin/export`

Export Types

Export Type	URL Example	Content
User data	/admin/export?type=users	User ID, name, student ID, email, role, and active status
Application data	/admin/export?type=applications	Application ID, user ID, module information, application role, date, status, and feedback

The platform supports CSV export for user data and application data, which is useful for reporting and administrative analysis.

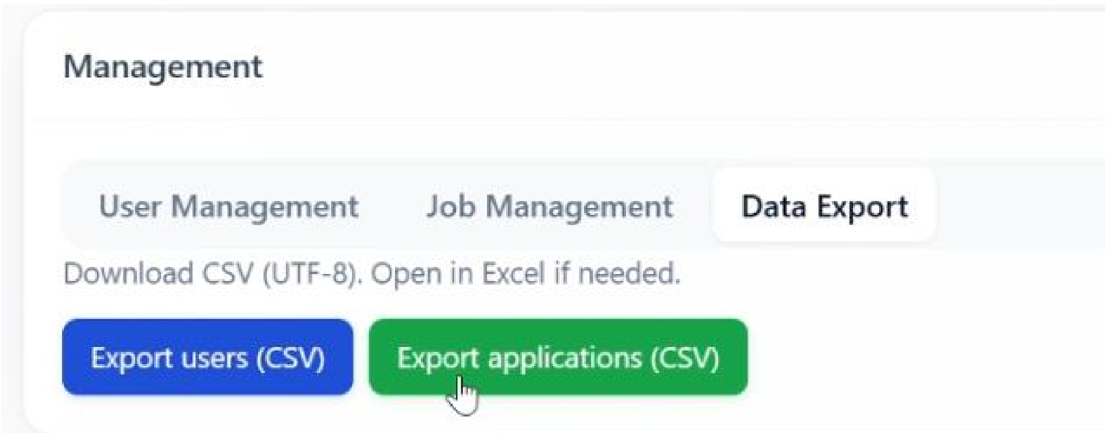
Steps

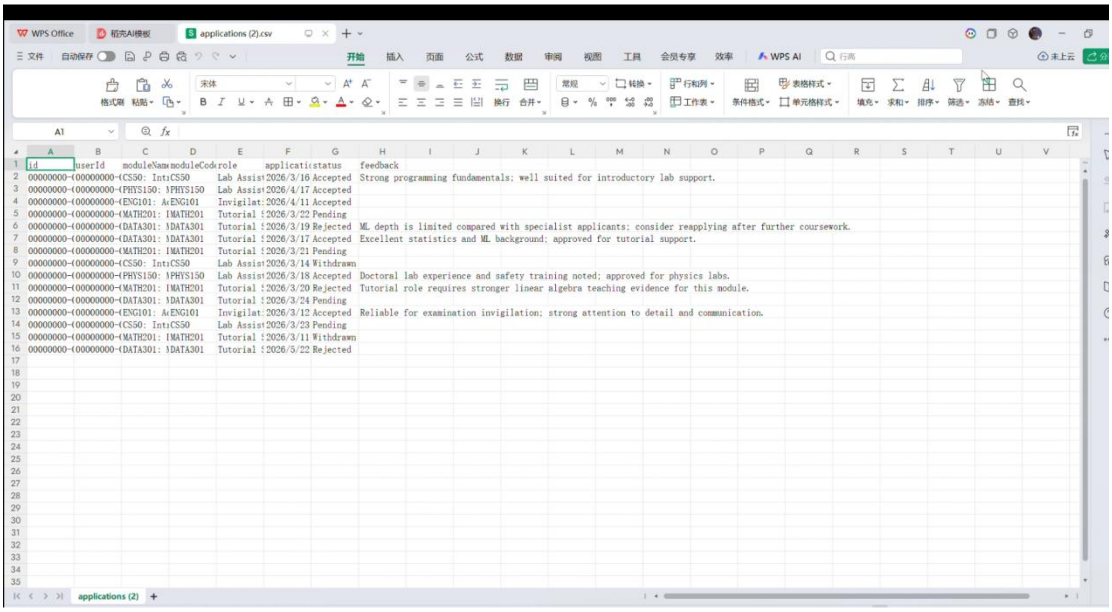
1. Log in as an Administrator.
2. Open the Administrator Dashboard.
3. Find the CSV export buttons or links.
4. Select the type of data to export.
5. Click the export button.
6. Download the generated CSV file.

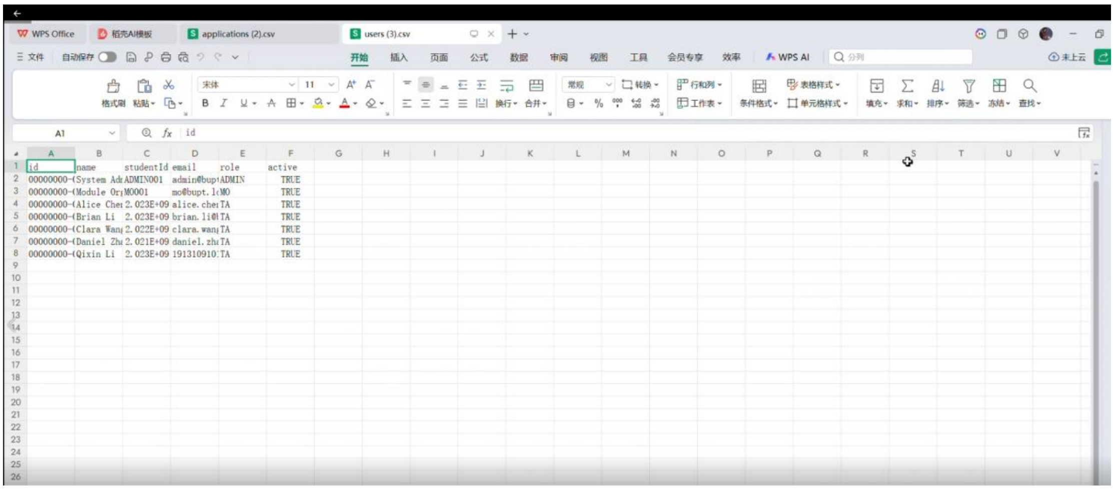
Expected Result

The system downloads a CSV file containing the selected system data.

Screenshot







4.7 TA Profile Overview

Purpose

The TA Profile Overview function allows administrators to browse all TA applicant profiles in read-only mode. This helps administrators review applicant information without logging in as each TA user.

Access

```
/admin/ta-profiles
```

CV download route:

```
/admin/cv?userId=userId
```

Main Information Displayed

Information	Description
Applicant information	Basic personal information of the TA applicant
Education background	Education history entered by the applicant
Skills	Applicant’s skills
Completed courses	Courses completed by the applicant
Available time	Applicant’s available working time

Information	Description
Uploaded CV	Allows the administrator to open or download the applicant’s CV

The feature map states that administrators can browse all TA profiles in read-only mode and open or download uploaded CV files from this page.

Steps

1. Log in as an Administrator.
2. Click **Profile** or open `/admin/ta-profiles`.
3. Browse the list of TA applicant profiles.
4. Check applicant information, education background, skills, courses, and available time.
5. Click **Open CV** if the applicant has uploaded a CV.
6. Return to the Administrator Dashboard when finished.

Expected Result

The administrator can review TA applicant profiles and CV files in a centralised read-only overview page.

4.8 AI Demo Page

Purpose

The AI Demo Page is designed for standalone AI testing and demonstration. It is different from the TA-facing AI functions because administrators manually enter candidate information and job requirements for testing.

Access

`/admin/ai-demo`

Available AI Tests

AI Function	Description
Skill matching	Compares candidate skills with job requirements

AI Function	Description
Missing skills analysis	Identifies missing skills for a candidate
Job recommendation	Provides job recommendation based on input information

The README describes `/admin/ai-demo` as an admin-only mock/demo AI scaffold for skill matching, missing skills, and job recommendation. The system also uses a mock provider by default, so the AI demo can be used without an external API key.

Steps

1. Log in as an Administrator.
2. Open the AI Demo page.
3. Enter candidate skills, job requirements, or job descriptions.
4. Select the AI test type.
5. Submit the input.
6. View the generated AI result.

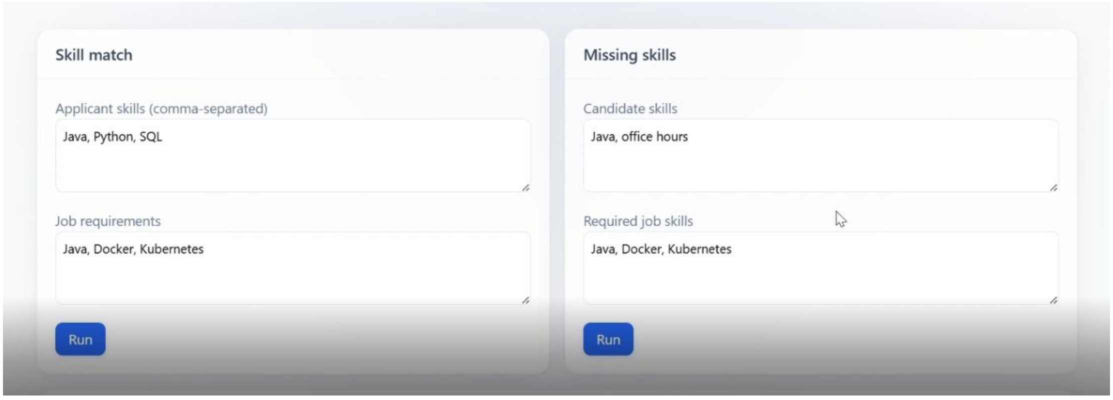
Difference from TA-Facing AI

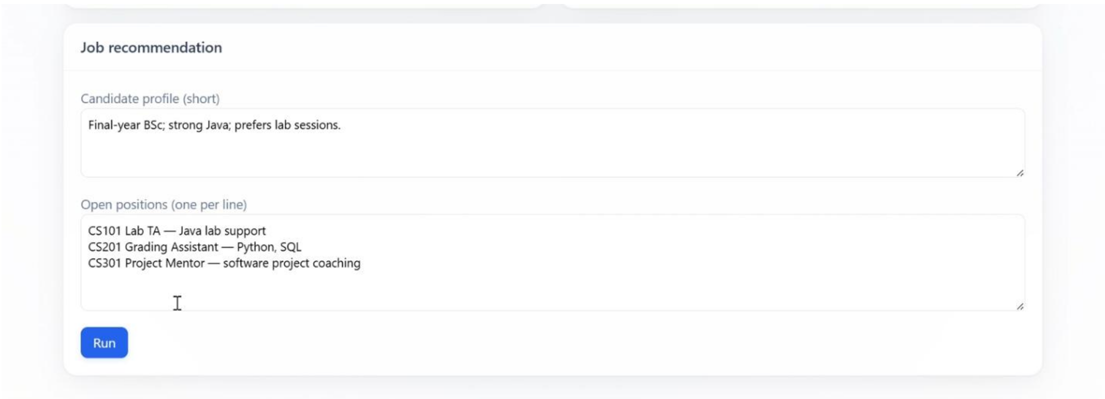
The TA-facing AI functions are embedded in the job list and job detail pages. They use the applicant's profile and job list information. In contrast, the Admin AI Demo Page is a standalone testing page where the administrator manually enters information. The result is used for system testing and AI provider verification.

Expected Result

The administrator can test AI-based skill matching, missing skills analysis, and job recommendation in a standalone demo environment.

Screenshot





可以，下面我把最后三部分直接写成：

5. Troubleshooting
6. Limitations and Notes
7. Conclusion

也就是说，原来的 **Role-Based Permissions** 不单独作为第 6 章了，相关内容我放进 **6. Limitations and Notes** 里面。

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5. Troubleshooting

This section describes common problems that users may encounter when using the TA Recruitment Management System. It also provides possible reasons and suggested solutions.

Problem	Possible Reason	Solution
The user cannot log in.	The email or password is incorrect.	Check the email address and password, then try again.
The account cannot	The account may have been	Contact the system

Problem	Possible Reason	Solution
log in even with the correct password.	deactivated by an administrator.	administrator for support.
A new user cannot register successfully.	Some information does not meet the validation requirements, such as student ID, email format, phone number, or password confirmation.	Check the error messages on the registration page and correct the invalid fields.
The applicant cannot apply for a job.	The personal profile is incomplete.	Complete all required profile information first, including personal details, education history, courses, availability, skills, and CV upload.
The Apply button is disabled.	The TA quota may be full, or the applicant does not meet the application rules.	Check the job details and application messages shown on the page.
The applicant cannot withdraw an application.	Only Pending applications can be withdrawn. Accepted or Rejected applications cannot be withdrawn.	Check the current application status on the My Applications page.
A job is not visible to TA applicants.	The job may still be saved as a draft.	The Module Organiser needs to publish the job before it becomes visible to applicants.
The MO cannot approve or reject an application.	The application may already have been processed.	Only Pending applications can be approved or rejected by the MO.
The uploaded CV cannot be opened.	The file may be missing, unsupported, or uploaded incorrectly.	Upload the CV again using a supported format such as PDF, DOC, or DOCX.
The AI recommendation is unavailable.	The applicant profile may be incomplete, or the skills field may be empty.	Complete the profile and enter relevant skills before using AI functions.
The admin cannot delete or deactivate a user.	The system prevents dangerous operations, such as deleting the only administrator or deactivating the current admin account.	Select another user account or keep at least one active administrator account.

The system includes several rules to protect the recruitment workflow. For example, only Pending applications can be withdrawn by applicants, MO users can only process Pending applications, and draft jobs are invisible to TA applicants until they are published.

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6. Limitations and Notes

This section explains important limitations and notes about the TA Recruitment Management System. Since the system is developed as a coursework prototype, some functions are simplified for demonstration purposes.

6.1 Coursework Prototype

The system is a lightweight coursework prototype for TA recruitment management. It is designed to demonstrate the main recruitment workflow, including TA registration, profile completion, job browsing, application submission, MO approval, administrator supervision, and AI-assisted features.

The system focuses on showing the complete workflow rather than providing a production-level recruitment platform.

6.2 Data Storage

The system uses plain JSON files to store data. User accounts, applicant profiles, job records, application records, and uploaded CV files are stored on the server side. There is no separate database in this prototype.

This design makes the system easier to run and demonstrate, but it may not be suitable for large-scale real-world deployment. In a production system, a database would normally be used to improve data management, reliability, security, and scalability.

6.3 Password Storage

In this coursework prototype, passwords are stored in plain text. This is acceptable only for demonstration and coursework purposes. In a real production system, passwords should be securely hashed, for example by using BCrypt or another secure password hashing method.

Users should not use real personal passwords when testing the prototype system.

6.4 AI Function Notes

The AI functions are designed to provide assistive recommendations and analysis. They do not automatically submit applications, approve candidates, reject candidates, or change application statuses.

The default AI provider is a mock offline provider, which means the system can demonstrate AI-related functions without requiring an external API key or network connection. The AI output should be treated as guidance only, and final recruitment decisions should still be reviewed by human users.

6.5 Password Recovery Limitation

The **Forgot Password** and **Reset Password** functions are demonstration placeholders. They are included to show where password recovery features would appear in the system, but they do not send real recovery emails or perform a full production-level password reset process.

6.6 Role-Based Limitations

Different users have different permissions in the system.

Role	Limitations
TA Applicant	Cannot approve applications, publish jobs, delete jobs, or access MO/Admin pages.
Module Organiser	Cannot force application status changes, manage users, delete jobs as an admin, export CSV data, or access the Admin AI Demo page.
System Administrator	Can supervise and manage the system, but job creation and publishing are mainly handled through the MO workflow.

These limitations help maintain role-based access control and prevent users from performing actions outside their responsibilities.

6.7 Application and Job Rules

The system follows several important business rules:

Rule	Description
TA quota rule	If a job has reached its TA quota, new applications cannot be submitted.
Application rule	A TA applicant can only have one active application for the same module when the status is Pending or Accepted.
Reapplication rule	If an application is Rejected or Withdrawn, the applicant may apply again if the job quota is not full.
Draft job rule	Draft jobs are not visible to TA applicants.
Published job rule	Published jobs become visible to TA applicants on the job list page.
Withdrawal rule	Only Pending applications can be withdrawn by applicants.
MO approval rule	MO users can only approve or reject Pending applications.
Admin override rule	Administrators can force-change application statuses when necessary.

These rules ensure that the recruitment workflow remains consistent across the TA applicant side, MO side, and administrator side.

---

7. Conclusion

The TA Recruitment Management System provides a complete online workflow for university Teaching Assistant recruitment. It supports the main recruitment stages from applicant registration to profile completion, job browsing, application submission, MO review, and administrator supervision.

For **TA applicants**, the system allows users to create accounts, complete personal profiles, upload CVs, browse published TA jobs, use AI-assisted recommendation and analysis functions, submit applications, track application status, view feedback, and withdraw Pending applications.

For **Module Organisers**, the system supports job and application management. MO users can view the MO dashboard, review incoming applications, open applicant CVs, view applicant profiles, approve or reject Pending applications, provide feedback, and create or publish TA job postings.

For **System Administrators**, the system provides higher-level supervision and management. Administrators can monitor recruitment statistics, check application status analysis, monitor TA workload, force-change application statuses, manage user accounts, inspect job records, export CSV data, browse TA profiles, and use the AI demo page for testing.

Overall, the system improves the efficiency and organisation of the TA recruitment process. It replaces scattered email-based and spreadsheet-based management with a more structured online workflow. By combining role-based access control with AI-assisted recruitment features, the system provides a more intelligent, transparent, and manageable platform for TA recruitment.